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Education

December 2024 to April 2026: Master of Science (Major Component of Research) Department of Mathematics, University of Moratuwa, Sri Lanka. Area: Applied mathematics with applications in Machine Learning, Thesis Title: Enhancing Classification Accuracy of Frequency Domain Data by Least Squares and Feature Engineering.

January 2020 to July 2024: Bachelor of Science of Engineering Honors Department of Earth Resources Engineering, University of Moratuwa, Sri Lanka. Obtained Upper Second-Class Honours.

Research Experience

January 2025 to date: Research Assistant Department of Mathematics, Faculty of Engineering, University of Moratuwa, Sri Lanka

Industry Experience

January 2023 to June 2023: Geotechnical Engineer (Intern) Sri Lanka Ports Authority, Colombo, Sri Lanka

- hands-on experience across geotechnical engineering, dredging, marine engineering, and electrical systems

Contributions to Research

Areas of interest: Applied Mathematics, Signal Processing, Machine Learning, Computer Vision, Data Science, Applied Probability and Statistics

Publications

- [1] Amirthavarman V., **Perera A. W. B. M.**, Madhuwantha G. L. S., Fernando W. A. M., Jayawardena C. L., & Wickrama M. A. D. M. G. (2024). Experimental synthesis of zeolites from pre-processed coal fly ash using the microwave irradiation method. *Proceedings of International Symposium on Earth Resources Management and Environment*, 170–178. <https://doi.org/10.31705/ISERME.2024.27>.
- [2] **Perera A. W. B. M.**, Premasiri M. H. S., Wedige M. S., & Fernando A. (2025). *Huber Monotonic Regression (HMR): A Smooth and Outlier-Resistant Framework for Nonparametric Regression with Modified Huber Loss*. <https://doi.org/10.2139/ssrn.5313703> (Preprint Available).
- [3] **A. W. B. M. Perera**, M. H. S. Premasiri, P. M. Edirisinghe, and W. A. M. Fernando, “Enhancing Fashion-MNIST classification accuracy: A study of regularised least squares and kernel-based approaches,” in *2025 9th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI)*, IEEE, Nov. 2025, pp. 1–6. doi: [10.1109/SLAAI-ICAI68534.2025.11318512](https://doi.org/10.1109/SLAAI-ICAI68534.2025.11318512).
- [4] **A. W. B. M. Perera**, W. A. M. Fernando, P. M. Edirisinghe, and M. H. S. Premasiri, “Random ReLU–Based Support Vector Machine Classifier for Frequency-Domain Data,” in *2026 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications*, IEEE, May. 2026, pp. 1–5 ([Accepted](#)).
- [5] **A. W. B. M. Perera**, W. A. M. Fernando, P. M. Edirisinghe, and M. H. S. Premasiri, “Gaussian ReLU Feature Expansion for Nonlinear Discriminant Analysis in High-Dimensional Classification,” in *2026*

8th International Conference on Cybernetics, Cognition and Machine Learning Applications, IEEE, October. 2026, pp. 1–6 (under review).

- [6] **Perera et al.**, *A Unified Framework for Scalable Least Squares Classification: Integrating Gaussian ReLU Features, Dimensionality Reduction, and Modified Huber Loss* (In preparation).

Certificates & Badges

- [1] [Machine Learning Specialization - DeepLearning.AI](#)
- [2] [Deep Learning Specialization - DeepLearning.AI](#)
- [3] [Generative AI for Data Scientists Specialization - IBM](#)
- [4] [Signals and Systems - MIT OCW](#)
- [5] [Matrix Calculus for Machine Learning and Beyond - MIT OCW](#)
- [6] [Introduction to Algorithms – MIT OCW](#)
- [7] [Introduction to CS and Programming using Python – MIT OCW](#)
- [8] [Deep Learning for Computer Vision – Stanford Online](#)
- [9] [Principles of Sensing for Autonomy – Stanford Online](#)
- [10] [Data Analysis with Python - IBM](#)

Projects Contributions

- [Fine-tuned Mistral 7b LLM for Mining and Mineral Engineering industry \(2024\)](#)
- [Build a Machine Learning Web App with Streamlit and Python \(2024\)](#)

Voluntary Work

- Served as a Member of Earth Resources Engineering Society, University of Moratuwa
- Served as a Member of the Mathematics Society, University of Moratuwa
- [Served as Department Coordinator for EXMO \(2023\)](#)

Professional Affiliations

- Graduate Student Member of the Institute to Electronic and Electrical Engineers (IEEE).
- Associate Member (AMIE) of Institution of Engineers Sri Lanka (IESL).
- Academic Member of Sri Lanka Association for Artificial Intelligence (SLAAI).

References

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